# Module 2 Lecture - Research Sources

Research Methods in Psychology

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# 1 Overview and Introduction

## 1.1 Objectives

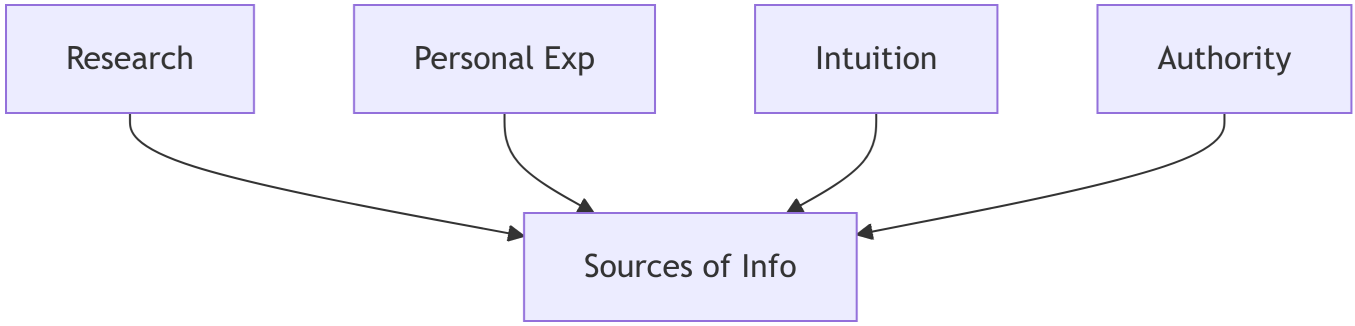
- Explain why all scientists, including psychologists, value research-based conclusions over beliefs based on experience, intuition, or authority.
- Locate legitimate research-based information and read it with a purpose.
- Be able to understand and appreciate why certain sources are valued above others and how we make this determination (i.e., *why* is research the best?)
- Be able to enumerate the reasons why a source is, or is not, a valid scientific resource, and why some sources may be of mixed validity
- Be able to explain how well a source follows the empirical, theory - hypothesis - data model
- Be able to recognize biases in your approach to literature review

## 1.2 Introduction

- We can gain _____ from many origins; this is true of both scientific and common sources and:
  - Scientific Sources:
    - Peer-reviewed Journal _____
    - Books and edited books
    - Scientific _____
    - etc.
  - Common sources:
    - _____ outlets (and lots of them!)
    - Newspapers
    - Think tanks (private "research")
    - "Well my friend said…"

📢 Discuss: What is the most common source of information you use for news, hobbies, interests, etc.?

- *But*, how do we _____ these sources - which ones are the "best" for drawing conclusions? (hint: research!)
  - Also, how do we read each of these _____ more critically?
- We'll compare 4 categories of sources:




# 2 Research vs. Personal Experience

## 2.1 Overview

- Personal experience is simply the _____ that comes at us from everyday life.
- This is when we learn from the _____ that occur to and around us.
- *Example:* I learn to walk carefully (like a penguin!) on ice because I have slipped in the past when walking normally. But does walking like a penguin actually work for people to avoid slipping?

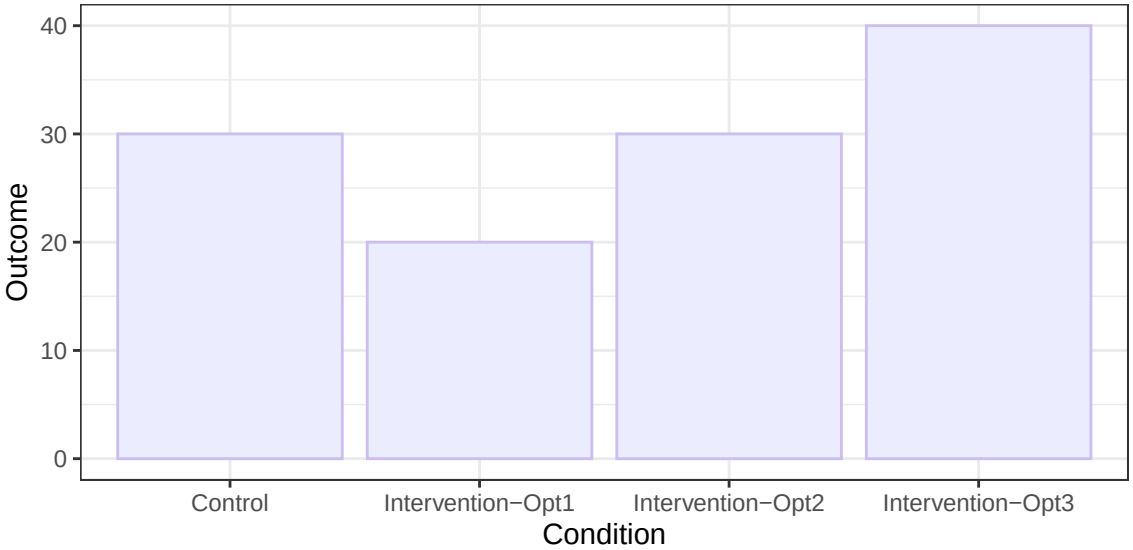
📢 Discuss: What is a goofy or funny life lesson you've learned that you still use to help make decisions?

- We often rely on personal experience to make _____, but this does not always mean it is the most reliable source of information for drawing big conclusion

> **Important**
>
> Please know that personal experience is valuable, and I'm not trying to disprove that!

## 2.2 No Comparison

- A comparison group is a key _____ between scientific research and personal experience. We must compare our study results to some other benchmark, whether that is a _____ group (no intervention) or another type of intervention.
- Without a comparison group we have three _____ outcomes:
  - Our effect is _____ than control (no intervention) or other interventions
  - Our effect is _____ to control or other interventions
  - Our effect is _____ than control or other interventions




- **Example:** A researcher performs a social intervention to reduce feeling of hostility towards an "out-group". It seems like the participants don't harbor any major negative feelings towards the out-group at the end of the study. A success!? Maybe…
  - Why are the _____ (potentially) wrong here? What other comparisons are possible?
- You may have some belief about the _____ of an intervention (from personal experience), but you won't have good _____ of

its "real" effect, without a comparison group.
  - Example: Neuropsychologist and decaf coffee

Discuss: Are there any habits or traditions you swear by, even if you don't know if they, work?

## 2.3 Experience is Confounding

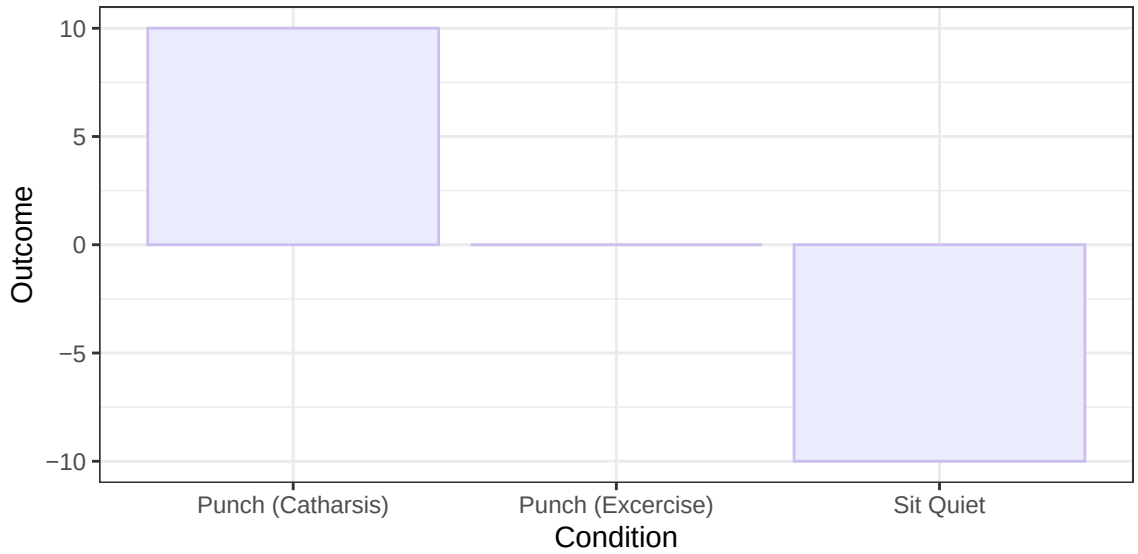
- Why did something happen? Confounds make it difficult to determine causation in our personal experiences. Confounds are _____ that may change how or why something happens.
- Example: A professor addresses you in a disappointed tone on the first day of class. You assume it is due to them just disliking students.

Discuss: Why are the conclusions (potentially) wrong here? What other explanations are possible?

- In our own lives it is difficult to impossible to isolate individual causes for certain outcomes, but in research, we have _____ and statistical controls for confounds

## 2.4 Research > Experience

- In research settings, we can better exert control for the _____ of confounds, and create comparison groups
- **Example:** Bushman's study (catharsis theory)




- Another tale of why scientific process > _____

## 2.5 Probability in Research

- Individual differences are _____ in theories!
  - There are _____/edge cases to all hypotheses and, as we mentioned in the last lecture, we cannot account for all possible scenarios with a single _____.
- Research is always probabilistic which means it try to capture the _____ of experiences or the net average of all cases
  - Which may mean your experiences may contradict the _____ of evidence in the research body - but that doesn't mean that your experience or the research consensus is "wrong"

> **Important**
>
> This is also why it's important that we don't try to assume that every person's experience is represented with research

- Connect to your previous _____ class: conclusions are based on p-value, which, in turn, are essentially a _____ of a certain outcome.
  - Just because we _____ a null hypothesis, does not mean we are *certain* of a result
- A single case does not fully invalidate a theory or hypothesis. Instead, it may

represent a unique case that differs in some way than the others. We must research _____ to understand why!

- *Example:* A medical doctor understands that a blood pressure drug works for _____ individuals with hypertension - but one of their patients isn't responding to it.

**?** Review: Which one of the following is *not* one of the reasons why personal experience doesn't work great for conclusions?

- (●) A) No comparison groups
- ( ) B) For control for confounds
- ( ) C) Personal experience may be driven by individual differences
- ( ) D) Personal experience lacks any evidence

Explanation:

# 3 Research vs. Intuition

## 3.1 Bias in Intuition

- Recall that intuition is often like a _____ or a "gut feeling". We may describe intuition as being "pre-scientific."
- Unfortunately, our intuition may be swayed by certain sources of _____ in how we think…

**Important**

Our brains are naturally biased, and there is no shame in admitting so! But it pays to know what biases you may suffer from

### 3.1.1 Convincing Narratives

- Have you ever met someone who could really _____ a story? Maybe a friend who is really good at making stuff up or a salesperson.

Discuss: Think of a story you believed from a friend, family member, or news, and later realized was wrong? (keep it lighthearted)

- Freud certainly could! Many of his early theories drew on observations that *seemed* reasonable _____, and his psychoanalytic theory is prevalent to this day.
  - This is a sort of "just-makes-sense" thought process, where we think - "well, naturally, it just _____ that way"
- However, this intuition to believe convincing narratives can be an incorrect _____ and may prevent us from fully exploring whether things are actually as we think they are
- *Example:* "Scared straight" program for scaring kids out of certain lifestyles

### 3.1.2 Availability Heuristic

- *Example:* Anyone afraid of shark attacks in the ocean?
- You may think that they happen often, but really, they are exceedingly _____. But because of popular media and news hype, the idea of shark attacks is much more readily **available** cognitively, we call this the availability heuristic.
- A heuristic is just a cognitive _____, it allows us to skip the process of thinking to come to conclusion.
  - However, in the scientific theory-data-hypothesis cycle, it threatens to _____ what is actually going on
- This heuristic often causes over- or under- estimations of certain situations _____

### 3.1.3 Present/Present Bias

- The present/present bias may be best understood as an extension of failing to account for _____ groups (see No Comparison)

  - If we don't _____ a possible outcome, do we fully account for it?
- *Example:* Remember Harlow's monkeys' third option…
- *Additional Example:* I ran in sneakers A and my run was great because of them! But what about sneakers B?
- This may also be understood as also reflecting a _____ to recall certain (critical) events, but not others

### 3.1.4 Confirmation Bias

- Confirmation bias is our tendency to _____ our already closely-held beliefs by selectively praising, criticizing, and noticing information that reinforces the belief
  - *Especially* relevant to situations like politics, where people have a great motivation to protect "their" _____
- This often causes _____ bias in evidence-gathering and is likely to lead to a single conclusion: "I'm right"

> **Important**
>
> Often one of the most insidious and difficult to overcome biases!

- *Example:*
  - Google search: "Evidence that the earth is flat" vs.
  - "Evidence of the earth's shape"

> Discuss: Write another pair of google searches that are likely to produce drastically different results

### 3.1.5 Bias Blind Spot

- Recall those last few examples of bias? _____ are just as susceptible to those same fallacies as anyone else.

  - This is called our bias blind spot

- It pays dividends to be _____ and curious towards possibilities, even when our intuition, experiences, and even research may say otherwise.
- Especially as scientists, there is a _____ to believe *we* are uniquely resistant to bias, *we aren't*
  - Want to eliminate bias? Prove it with sound science, methods, and reporting

**?** I think herbal supplements may offer some benefit for dementia, in part because I just saw several online ads as I was doomscrolling. This may be an example of...

- (●) A) Confirmation bias
- ( ) B) Availability heuristic
- ( ) C) Present/present bias
- ( ) D) Bias blind spot

Explanation:

## 3.2 Intuitive Thinking vs Personal Experience vs Scientific (Empirical) Thinking

- Taken together, there are lots of ways that intuition and _____ experience can let us down in the general conclusions we _____
- In science, we are interested in using *empirical* principals to guide our thinking and writing, as to capture the overall _____ in data and phenomena (remember, science is probabilistic)
- Simply "conducting research" does not _____ protect us from biases - we must be keenly mindful of the risks on biases in our research processes, starting with literature reviews

**Important**

Our intuition and personal experiences are not useless! But we have to set them aside, best we can, when conducting research

# 4 Research vs. Authority

## 4.1 Faults in Appealing to Authority

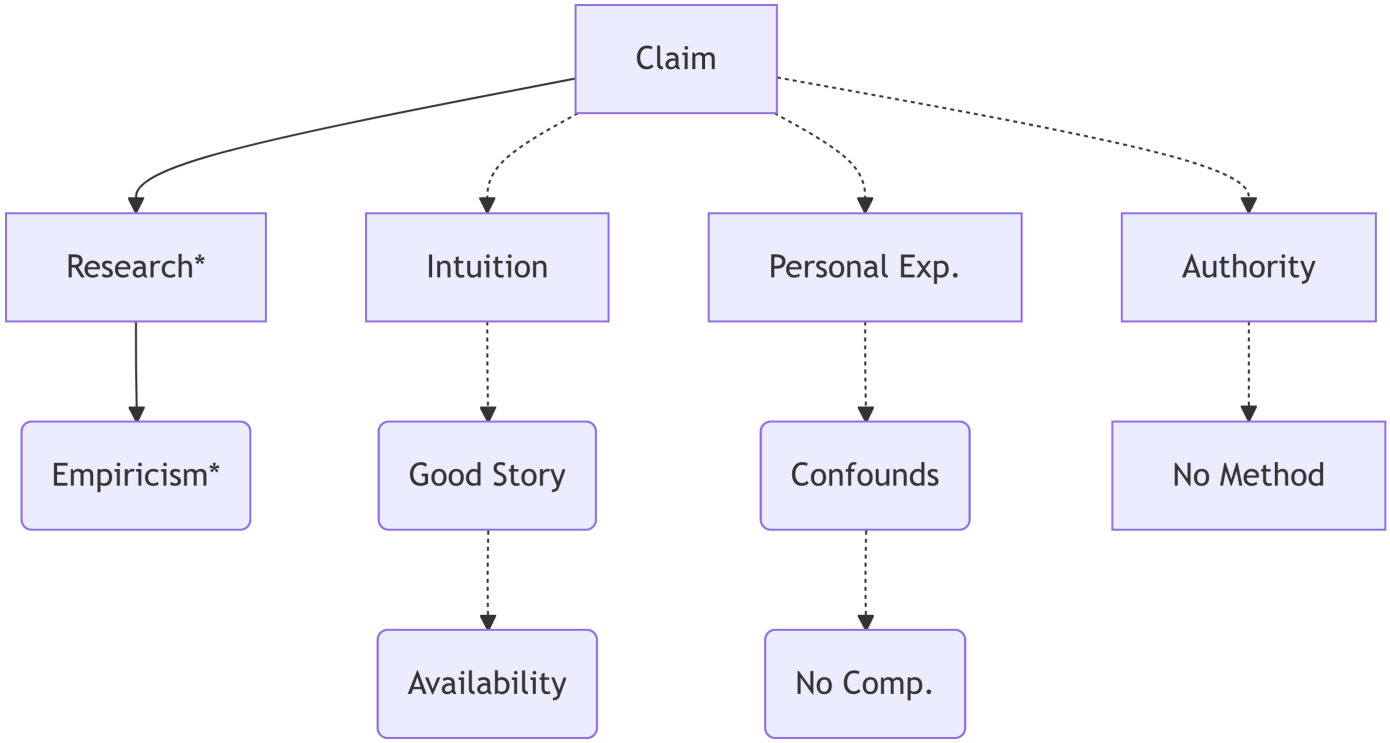
- We tend to _____ many sources of authority throughout our lives

Discuss: Who are the authority figures you trust in your life?

- However, we ought to distinguish between trusting a _____ for their perceived _____ and trusting *a process* for producing sound claims
  - Why do we trust journal articles? It isn't just because the researchers are simply _____, but rather, that the peer-review process is well set up to prevent poor science from slipping through
- Therefore, we must _____ *the method* by which a person comes to a conclusion, rather than basing our opinion of the person. This is the purpose of written articles, to interrogate the *how*, rather than the *who*

Discuss: I look up to my research mentor, who has worked hard to design and execute a careful, rigorous study on relationships in teenage couples. When talking to a friend, I state a claim my mentor made, and say that they're an expert on the topic - is this an appeal to authority?

### 4.2 Flow Chart of Claims




## 5 Examining Research Sources

### 5.1 Types of Articles

- Original empirical journal articles: Some form of _____ study in which a scientific process and analysis were conducted. These are often some type of novel observation or _____. Usually contains some form of introduction, methods, results, and discussion.

> **Important**
>
> Orignal empirical journal articles are the 'gold standard' for primary information about a topic or hypothesis

- Literature review journal articles: A comprehensive literature review that _____, synthesizes, and compounds the many available empirical studies in a specific research area. Often, comes across as an extended introduction section.
  - Also, a great starting point when doing a literature review for your own _____!

- Meta-analysis: An extension of a literature review articles that calculates a _____ effect size from the surveyed research studies. A more quantitative version of a classic literature review.

## 5.2 Types of Books

> **Important**
>
> Books are important, and tend to be great resources, but are usually considered somewhat less respected than journal articles

- Scientific books: a full-length scientific book written by the same single or multiple authors throughout. Focuses on a single _____, and functions as a sort of extended literature review.
- Edited books a collaborative effort in which many experts compose _____ on specific topics. These are usually not peer-reviewed to the same extent as journal articles - but the authors are usually some of the most respected researchers in an area. Be careful of appeals to _____ because of this!
  - These are popular among applied practitioners of psychology ( _____ ), because it allows them to keep up with developments and ideas in a nice range of areas.

## 5.3 Finding & Reading Research

- We will cover this in a separate workshop/presentation! This is a critical skill to develop as an early researcher, and is fairly complicated.
- We will also cover how to determine the sections of articles, and what questions you should ask of yourself in each section.

## 5.4 Scientific Journalism's Role

- Empirical articles will *always* be the _____ way to gain scientific knowledge. Edited books, literature reviews, meta-analyses, etc. may all be good _____ sources as well.

- However, popular scientific journalism may help you find new, findings to your interests - just don't cite it directly. Instead, go to the original articles.

> **Important**
>
> Just like intuition, scientific journalism has it's place in supporting research, you just want to avoid treating it as a primary source.

# 6 Conclusion

## 6.1 Recap

- In this presentation, we went over the many ways we come to information and conclusions, and how many sources fall short of empirical research
- We discussed the common pitfalls in our intuition, personal experience, and authority, and how they relate to the relative strength of empirical research
- We also discussed some of the types of scientific sources, with various formats providing different levels of validity, rigor, and specificity
- In conclusion, we re-affirmed Psychology as focused on a high degree of rigor and strong standards to support our empiricism

# Key Terms

### A

**authority** In research, the sense that something has a perceived expertise, based on their status 11
**availability heuristic** The tendency to readily recall or believe information because it is available cognitively 8

### B

**bias blind spot** Our difficulty in recognizing our own biases 10

### C

**comparison group** A relative group we compare results to, in order to move past personal experience 4

**Confirmation bias** The tendency to seek out information only that supports our previously held beliefs 9
**Confounds** Things that make more complicated our explanations for relationships between two or more things 5
**control** In studies, it is our ability to carefully curate the context and surrounding of phenomena, for more rigorous study 5
**convincing narratives** A tendency in intuition to believe in stories that just seem true 8

**E**

**Edited books** A collaborative book effort by several authors to cover a broad area 13

**H**

**heuristic** A cognitive shortcut our brain takes to decrease processing power 8

**I**

**Individual differences** A description of how there is natural variance and unique traits in each person, making generalization not effective when applied to individuals 6
**intuition** Our gut feeling about something 7

**L**

**Literature review journal articles** Articles that review and summarize a body of literature in an area 12

**M**

**Meta-analysis** A type of review that uses statistical averaging 13

**O**

**Original empirical journal articles** The 'gold standard' articles that contribute novel research 12

**P**

**p-value** In statistical hypothesis testing, the probability that results occurred due to random chance, assuming the null hypothesis is true 6
**Personal experience** Information gathered via our life experiences 3
**present/present bias** The tendency to not consider an option or outcome 8
**probabilistic** The trait of empirical research that suggest we try to capture a general experience and that our results are, in some amount, based in chance 6

**S**

**Scientific books** A full-length book that focuses on a single topic, like an extended literature review 13

*The instructor-provided glossary may not include all terms worth memorizing, make sure you consider using the vocabulary list in you book and your own judgment to make sure you have all relevant terms*